



S.P.M College, Udantpuri

Bachelor Of Computer Application (BCA)

Part -1 (Paper-1)

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Computer Fundamental

COMPUTER SOFTWARE

❖ Language Translators

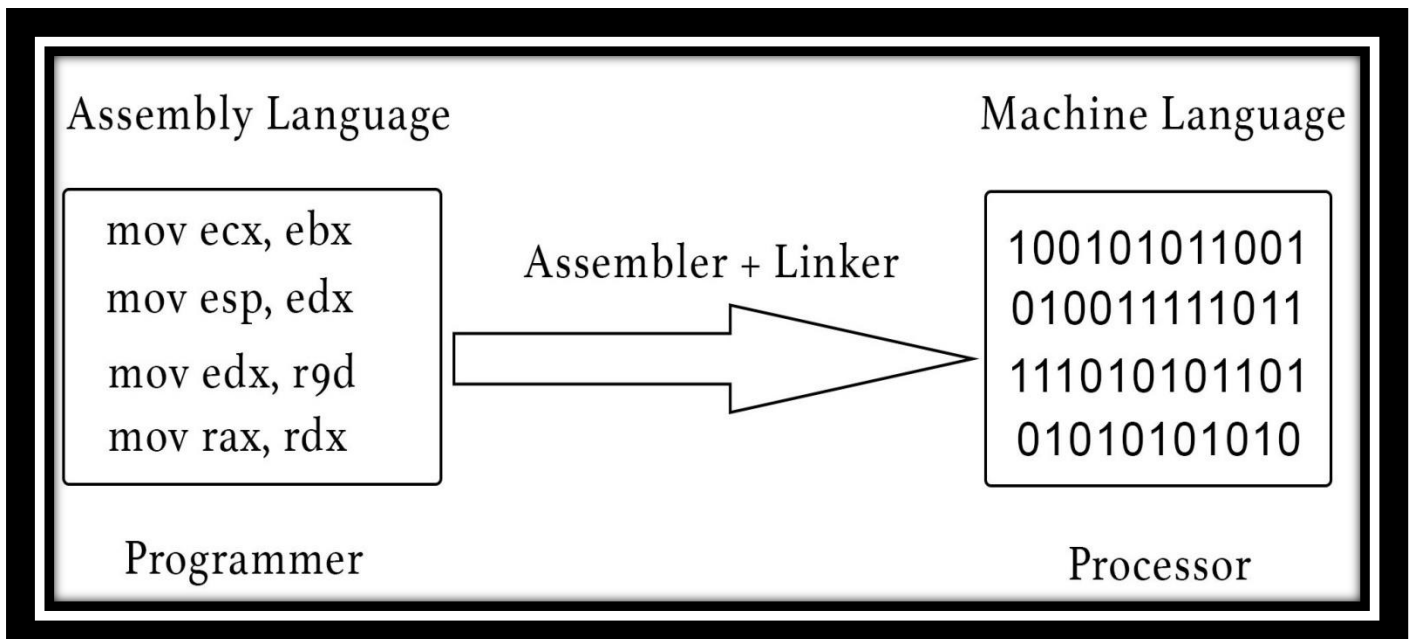
- It's a type of System software.
- These are such programs, that translate programs written in various programming languages(C++,Java,python,..) into the machine language of the computer.
- This translation is necessary because the computer can only follow programs written in its machine language.

Three types of translators

- i) Assembler
- ii) Compiler
- iii) Interpreter

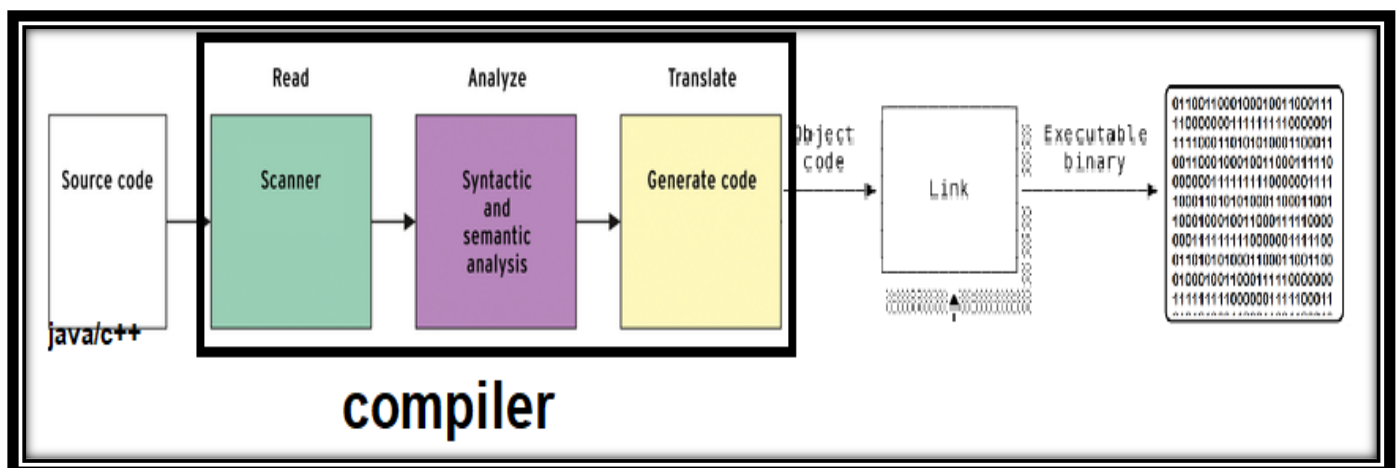
i) Assembler

- Assembly language program must be converted into its equivalent machine language program before can be executed.
- This translation is done with the help of a translator which is known as assembler.
- So Assembler translate convert assembly language into machine language.



ii) Compiler

- Compiler compile/convert/translate High level language(HLL) program into machine language program before can be executed.
- Compiler first scan all Source program/code, second analyze source code(syntax,...) and again all source program (not line by line) convert into machine language.
- During the translation the source program is only translates not executed.
- Every High level language (HLL) require different – different type of compiler. (e.g. – Java – compiler a, C- compiler b, C++ - compiler c, Python – compiler d,.....)
- A compiler can also detect & indicates the syntax errors during the compilation process but cannot able to detect logical errors.



iii) Interpreter

- Interpreter convert High level language into machine language.
- It takes one statement of high level language, translates into machine language & immediately executes the resulting machine language instructions.
- The main difference between compiler & interpreter is that compiler can translates the entire code but not involve in execution.
- Interpreter translates & executes a high level language program statement-by- statement / line by line.
- The main advantage of interpreter is that interpreter makes it easier & faster to correct programs.
- The main disadvantage is that interpreter is slower than compilers when running a finished program.



❖ Difference B/w to Compiler, Interpreter & Assembler

PARAMETERS	COMPILER	INTERPRETER	ASSEMBLER
Conversion	It converts the high-defined programming language into Machine language or binary code.	It also converts the program-developed code into machine language or binary code.	It converts programs written in the assembly language to the machine language or binary code.
Scanning	It scans the entire program before converting it into binary code.	It translates the program line by line to the equivalent machine code.	It converts the source code into the object code then converts it into the machine code.
Error Detection	Gives the full error report after the whole scan.	Detects error line by line. And stops scanning until the error in the previous line is solved.	It detects errors in the first phase, after fixation the second phase starts.
Code generation	Intermediate code generation is done in the case of Compiler.	There is no intermediate code generation.	There is an intermediate object code generation.
Execution time	It takes less execution time comparing to an interpreter.	An interpreter takes more execution time than the compiler.	It takes more time than the compiler.
Examples	C, C#, Java, C++	Python, Perl, VB, PostScript, LISP, etc...	GAS, GNU

❖ Utility Software

- Utility Software is a type of system software that helps to maintain, manage, and optimize the computer system.
- It performs tasks such as protecting the computer, managing files, and improving system performance.
- Utility software helps the computer run smoothly and efficiently.

Functions of Utility software

- a. System Maintenance** → Keeps the computer running properly.
- b. Security** → Protects the system from viruses and malware.
- c. File management** → Helps in organizing and managing files.
- d. Performance improvement** → Improves system speed and efficiency.

Image - <https://sun-infotech.in/wp-content/uploads/2023/04/utilityf-1-1024x664.jpg>

Example of Utility Software

1. Antivirus → Quick Heal, Avast,....
2. Disk Cleanup
3. Disk defragmenter
4. Backup utility
5. File compression tools → WinZip, WinRAR,...

❖ Device Driver

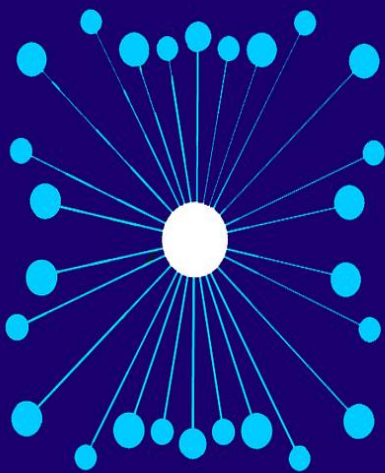
- A device driver is a type of system software that allows the operating system to communicate with hardware devices connected to the computer.
- A device driver acts a translator b/w operating systems and

the hardware device.

❖ **Centralized, decentralized, and distributed processing**

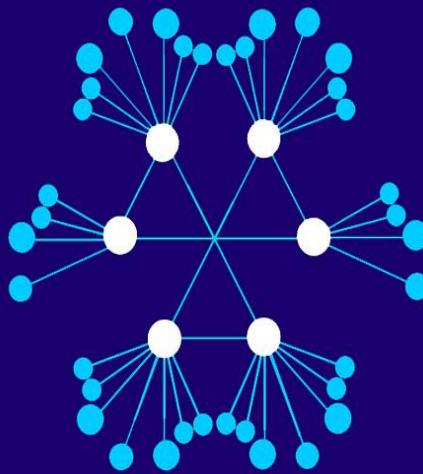
There are three main approaches to organizing computing resources and data processing in networks and systems.

Centralized vs Decentralized vs Distributed Network: An Overview



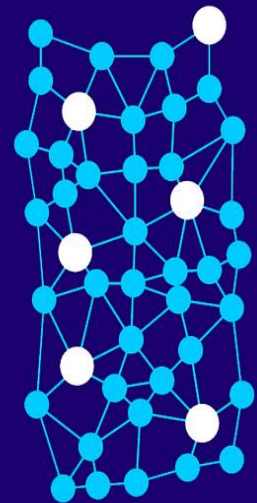
Centralized Network

All the nodes are connected under a single authority



Decentralized Network

No single authority server controls the nodes, they all have individual entity



Distributed Network

Every node is independent and interconnected with each other

1. Centralized Processing

- All data processing occurs (होना) at a single central location, typically a mainframe or server, while users access the system remotely.
- Examples: Main database of large companies, server of a website (amazon, job website, BGMI, TV Channel...)

➤ **Characteristics**

- A central server: All data and processing takes place on a single central server.
- Centralized control: All operations are controlled from a single location/node.
- High dependency: Users fully depend on the central system for

availability and performance.

➤ **Advantages:**

- Easier to implement security and enforce policies.
- Easy backup and data recovery processes.
- Reduced redundancy in data storage.

➤ **Disadvantages:**

- Single point of failure (if the central server fails, the entire system halts).
- Performance less as the system scales with more users.
- Increased latency(विलंब) for geographically dispersed users.

2. Decentralized Processing

- Multiple independent systems or nodes perform data processing, without the need for third parties/ central nodes.
- Example: Branch offices with their own servers managing local operations (Bank,Railway,SSC job site,UIDAI....)

➤ **Characteristics:**

- Systems operate independently of each other.
- Data and control are distributed across various locations.

➤ **Advantages:**

- Re-silient to failures (failure in one node doesn't affect the others).
- Customization possible at each node based on specific requirements.
- Reduced load on a central system.

➤ **Disadvantages:**

- Difficult to maintain consistency across nodes.
- More complex security and management.
- More expensive / Cost.

3. Distributed Processing

- Processing is distributed across multiple systems or nodes that communicate and collaborate to perform tasks.
- This is often seen in modern computing environments.
- Example: Cloud computing, Blockchain, College Computer Lab, grid computing systems.

➤ Characteristics:

- Nodes work together and share processing tasks.
- High interconnectivity and coordination between systems.

➤ Advantages:

- Scalability (adding more nodes enhances/increase capacity).
- Fault tolerance and reliability (failure of one node doesn't collapse the entire system).
- Optimized resource utilization.

➤ Disadvantages:

- Complex design and implementation.
- Increased communication overhead between nodes.
- Very expensive/ Costaly.

Parammeters	Centralized Systems	Decentralized Systems	Distributed Systems
Definition	Single central server controls and manages all operations.	Multiple nodes with independent control, no central authority.	Multiple interconnected nodes working together as a single system.
Control	Centralized control with a single point of management.	Distributed control, each node operates independently.	Shared control, nodes collaborate to achieve common goals.
Single Point of Failure	High risk; if the central server fails, the whole system fails.	Reduced risk; failure of one node does not impact the entire system.	Reduced risk; designed for fault tolerance and redundancy.
Resource Utilization	Central server resources are heavily utilized.	Resources are spread across multiple nodes.	Efficient resource sharing across nodes.
Performance	Can be high initially but may degrade with increased load.	Generally good, performance improves with more nodes.	High performance due to parallel processing and resource sharing.
Management	Easier to manage centrally.	More complex, requires managing multiple nodes.	Complex, requires coordination and management of many

Parameters	Centralized Systems	Decentralized Systems	Distributed Systems
			nodes.
Latency(विलंब)	Lower latency, as operations are managed centrally.	Can vary, depends on the distance between nodes.	Potentially higher latency due to network communication.

❖ Management information system (MIS)

1. Management: Art of accomplishing goals via and in collaboration with members of formally established groups. The following are managerial responsibilities:

- Planning
- Organizing
- Staffing
- Directing
- Controlling

2. Information: Data with a context and a meaning, where data is unprocessed information about an entity (entity is the object of interest)

3. System: A group of interconnected elements with a distinct boundary cooperating to accomplish a single objective.

- It is used for processing data. In an institution, employees, managers, and staff access MIS. Employees use MIS for day-to-day operations, to print invoices, bill payments, or performance reviews.
- An MIS collects data from various online systems to support management decision-making, analyses the information, and reports data.

Management Information System Components

- People, Data, Business processes, Hardware, Software



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